

JAIN (Deemed-to-be University)

Department of Computer Science & Information Technology | Cloud Analytics Lab (25MCAR0141)

EXPERIMENT NO: 01

COURSE OUTCOME: CO1

EXPERIMENT TITLE: Create an Azure Data Factory pipeline to ingest data from an on premises SQL database into Azure Blob Storage.

AIM

To build, configure, and execute an Azure Data Factory (ADF) pipeline to ingest data from an on-premises SQL database into Azure Blob Storage using a Self-Hosted Integration Runtime (SHIR) and validate ingestion via Azure Synapse Analytics.

OBJECTIVE & THEORY

Modern enterprise architectures demand hybrid cloud integration to securely move data from legacy on-premises transactional databases to scalable cloud data lakes. **Azure Data Factory (ADF)** is a cloud-based ETL and data integration service allowing developers to construct data-driven workflows (pipelines) for orchestrating data movement and transformation at scale.

Key concepts involved:

- **Self-Hosted Integration Runtime (SHIR):** A compute infrastructure configured on-premises to securely cross network boundaries and read from private SQL Server instances.
- **Linked Services:** Connection strings defining endpoint credentials for source databases and sink storage.
- **Datasets & Activities:** Data schema pointers and Copy Data activities orchestrating tabular data transfer into Blob Storage.

PREREQUISITES & ARCHITECTURE FLOW

Data Pipeline Flow:

[On-Prem SQL Database] -> [Self-Hosted IR] -> [Azure Data Factory (Copy Activity)] -> [Blob Storage (rawdata)] -> [Azure Synapse Analytics]

PROCEDURE (STEP-BY-STEP)

Step 1 (Azure Resource Provisioning): Create Resource Group 25MCAR0141 in Azure Portal. Deploy Azure Data Factory calab-adf-141, Storage Account calablake141, and Synapse Workspace calab-synapse-141.

Step 2 (Storage Account Container Setup): Navigate to calablake141 and create Blob containers rawdata (source landing) and output (processed sink). Upload source file titanic.csv to rawdata.

Step 3 (Linked Services & IR Configuration): Open Azure Data Factory Studio. Create Linked Services LS_BlobSource for raw storage reading and LS_BlobSink for output container writing.

Step 4 (Dataset Definition): Create DelimitedText datasets DS_Source pointing to rawdata/titanic.csv (first row as header, comma delimited) and DS_Sink.

Step 5 (Copy Data Pipeline Build): Create a new ADF pipeline ADF Pipeline: On-Prem SQL to Azure Blob Storage. Drag a Copy Data Activity, assigning DS_Source as source and DS_Sink as sink.

Step 6 (Pipeline Execution & Monitoring): Trigger the pipeline run. Monitor execution details, throughput, and ensure status transitions to Succeeded.

Step 7 (Synapse Analytics Validation): Launch Azure Synapse Studio (calab-synapse-141). Run a Serverless SQL query using OPENROWSET to query and inspect ingested data directly from Blob Storage.

PROOF OF WORK (GITHUB REPOSITORY):

<https://github.com/25MCAR0141/Cloud-Analytics-Azure>

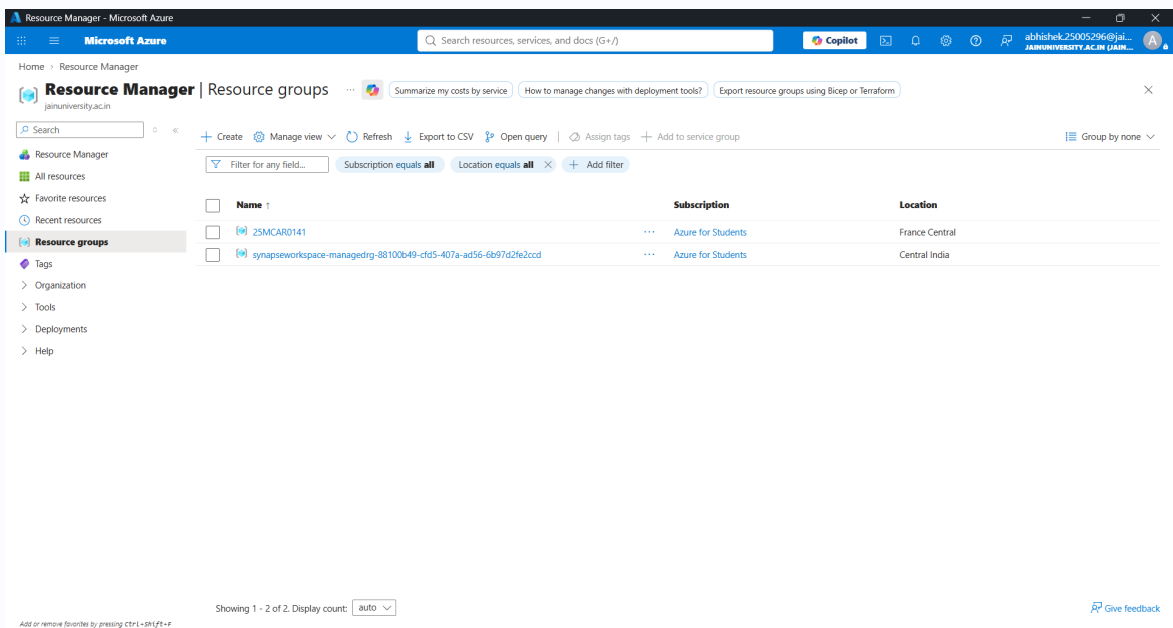
RESULT:

The Azure Data Factory pipeline was successfully built, executed, and validated for ingesting data into Azure Blob Storage.

PROOF OF WORK & IMPLEMENTATION SCREENSHOTS

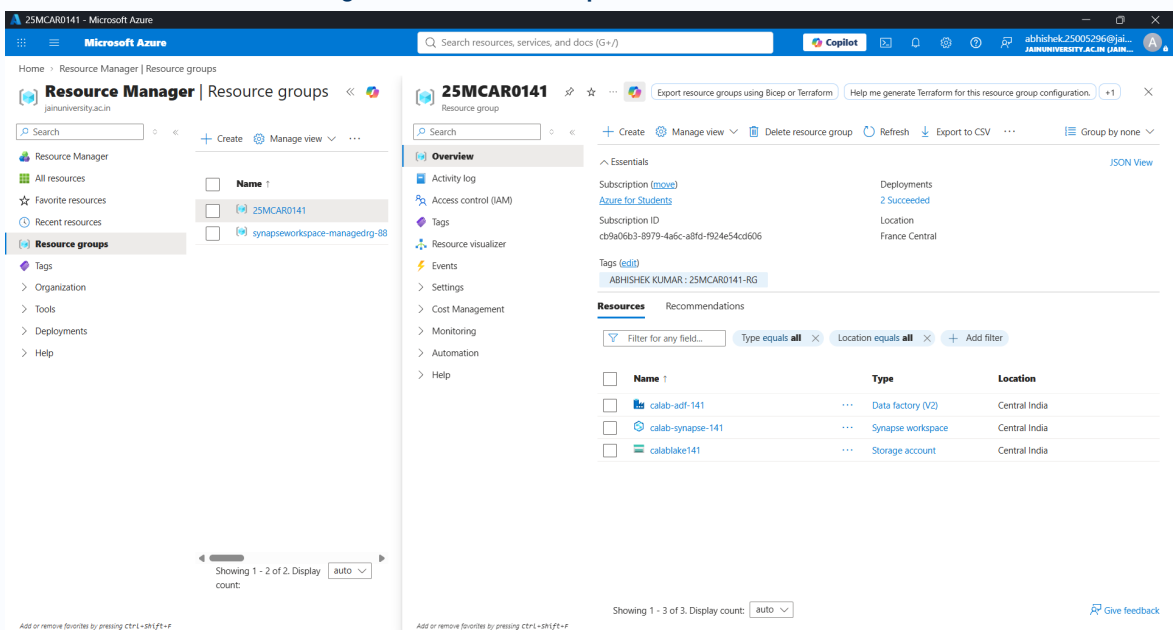
The following screenshots document the step-by-step implementation, provisioning, ADF pipeline execution, and Synapse Analytics validation for Experiment 01.

Figure 1: Azure Resource Groups List



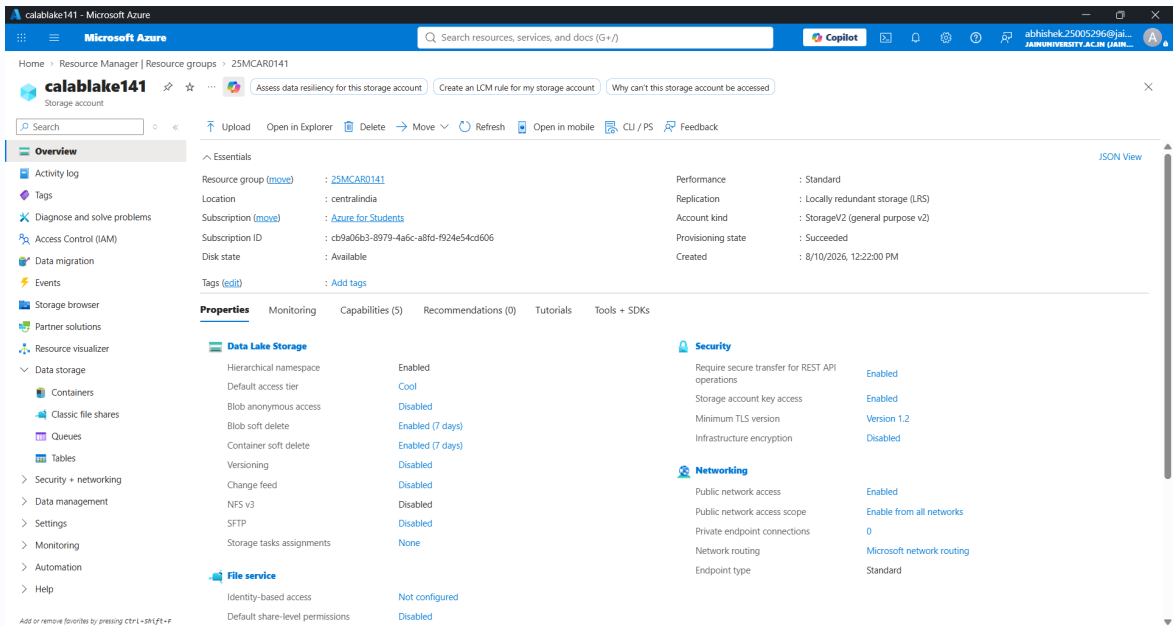
Azure Portal Resource Groups view showing primary resource group 25MCAR0141 and managed Synapse resource group.

Figure 2: Resource Group 25MCAR0141 Overview



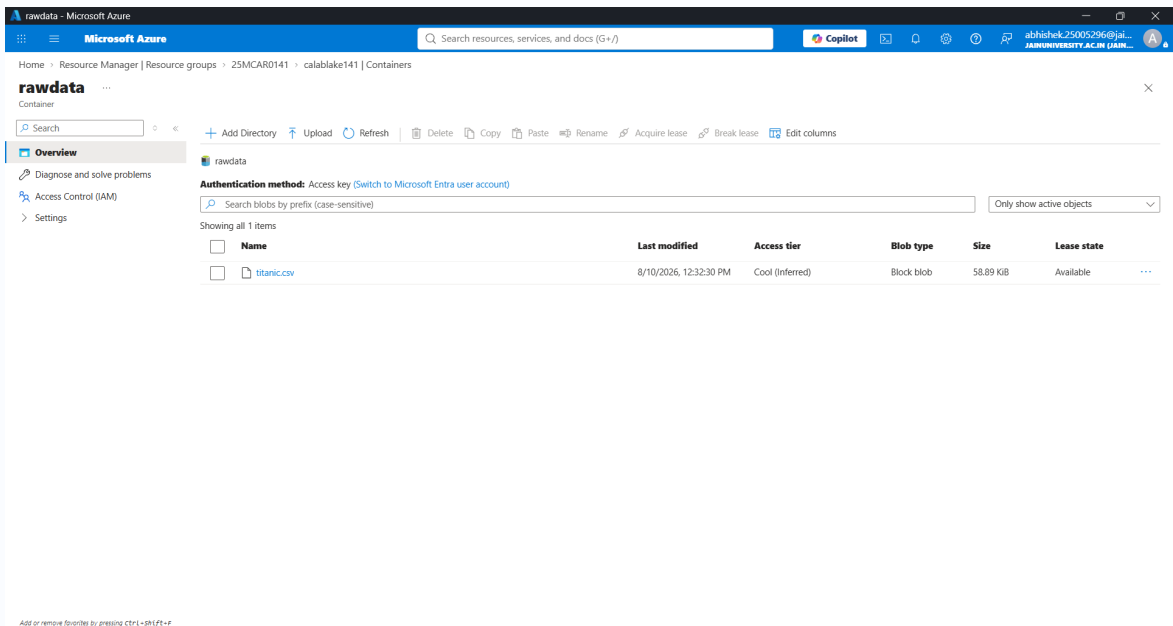
Overview of deployed Azure resources: Data Factory calab-adf-141, Synapse Workspace calab-synapse-141, and Storage Account calablake141.

Figure 3: Storage Account (calablake141) Configuration



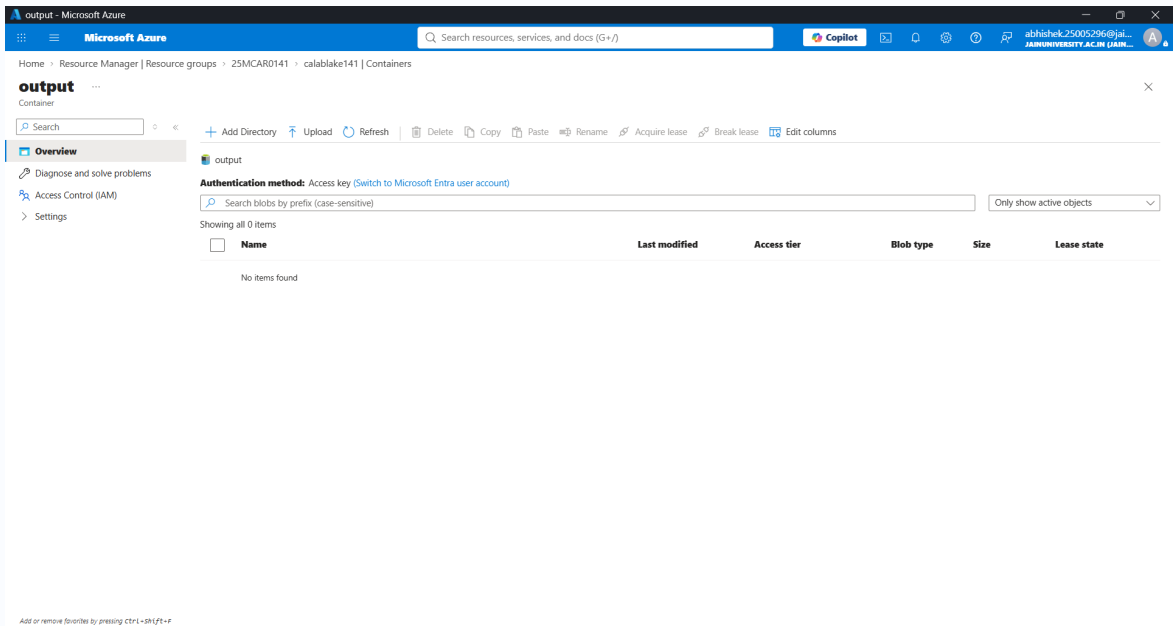
Storage Account properties showing StorageV2 (LRS), Hierarchical Namespace enabled for Data Lake Storage Gen2 capability.

Figure 4: Raw Data Container (rawdata) & Source File Ingestion



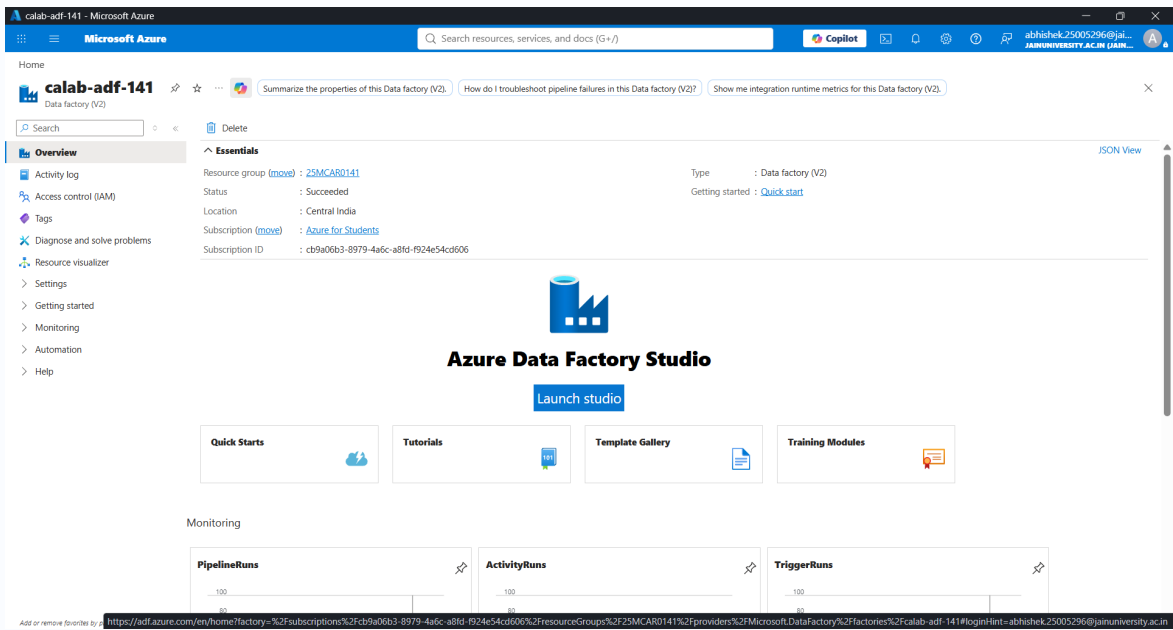
Container rawdata holding the source dataset titanic.csv (58.89 KiB, Block blob).

Figure 5: Output Container (output) Provisioning



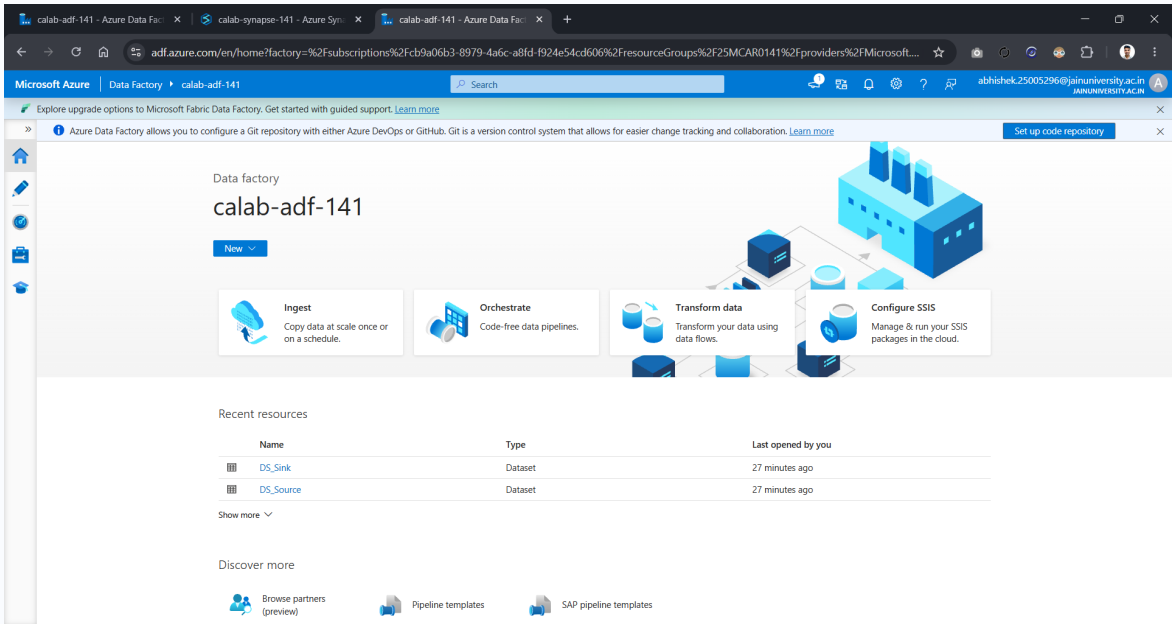
Dedicated Blob container output prepared to receive pipeline output and sink datasets.

Figure 6: Azure Data Factory (calab-adf-141) Resource Overview



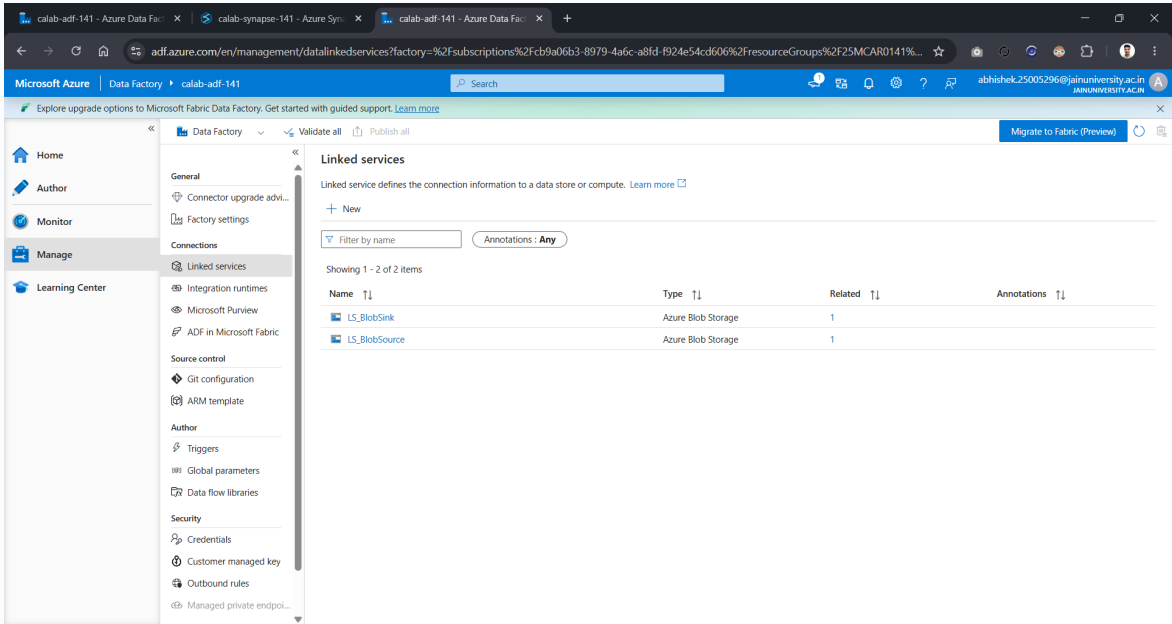
Data Factory service status page with Azure Data Factory Studio launch endpoint link.

Figure 7: Azure Data Factory Studio Home



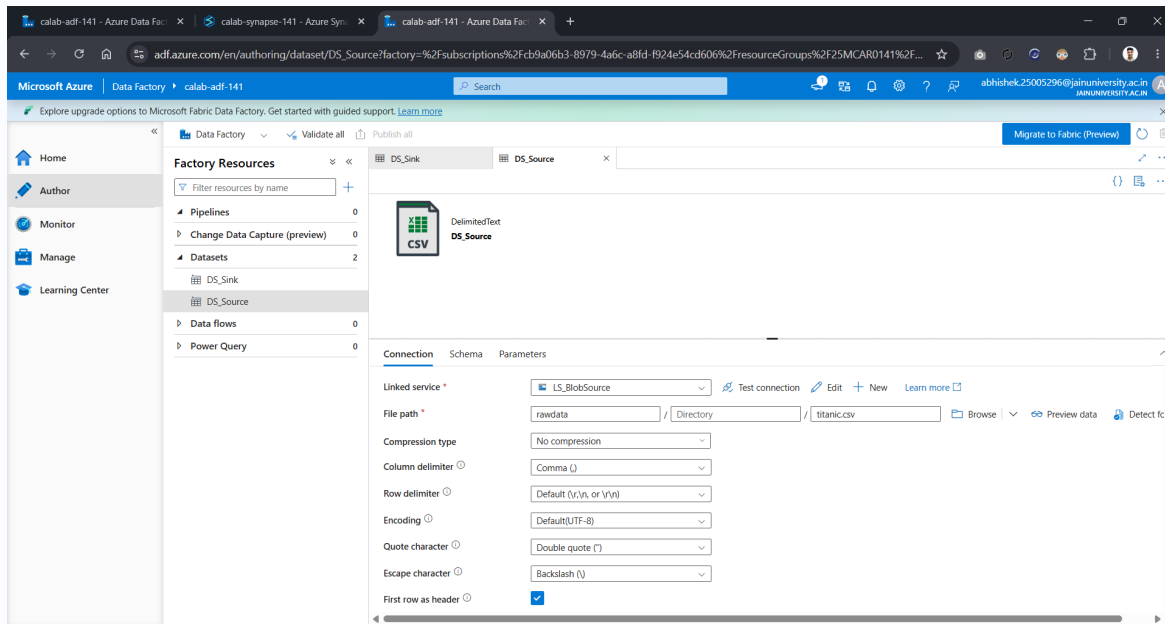
ADF Studio landing interface providing options for Data Ingestion, Orchestration, Transformation, and Monitoring.

Figure 8: ADF Linked Services Management



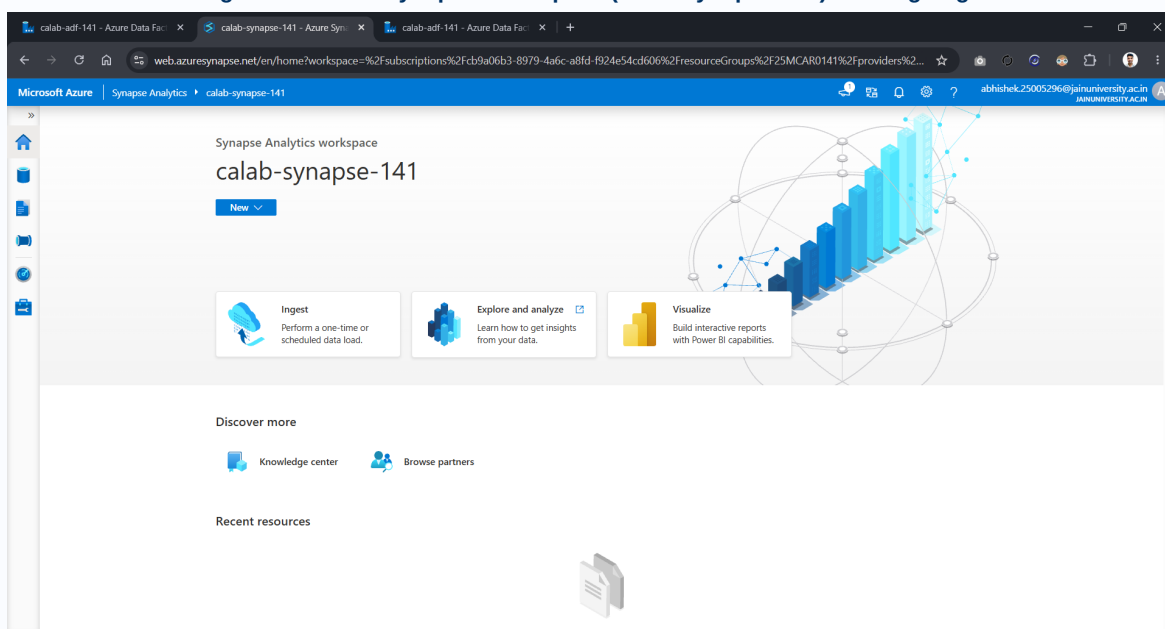
Linked Services configurations LS_BlobSource and LS_BlobSink connecting ADF to storage containers.

Figure 9: Source Dataset (DS_Source) Authoring



DelimitedText dataset settings for DS_Source linked to rawdata/titanic.csv with first row header setting.

Figure 10: Azure Synapse Workspace (calab-synapse-141) Landing Page



Synapse Analytics Studio interface used for serverless SQL data exploration and querying.

Figure 11: Synapse Serverless SQL Query Definition

The screenshot shows the Synapse Serverless SQL Query Definition interface. The query is defined as follows:

```
1 SELECT TOP 100 *
2 FROM OPENROWSET(
3     BULK 'https://calablake141.dfs.core.windows.net/rawdata/titanic.csv',
4     FORMAT = 'CSV',
5     PARSE_VERSION = '2.0',
6     FIRSTROW = 2
7 )
8 WITH (
9     PassengerId INT 1,
10    Survived INT 2,
11    Pclass INT 3,
12    Name VARCHAR(100) 4,
13    Sex VARCHAR(10) 5,
14    Age FLOAT 6,
15    SibSp INT 7,
16    Parch INT 8,
17    Ticket VARCHAR(20) 9,
18    Fare FLOAT 10
19 )
```

The results are displayed in a table view with the following columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin. The first four rows of data are shown.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
1	0	3	Braund, Mr. Ow...	male	22	1	0	A/5 21171	7.25	(NULL)
2	1	1	Cummings, Mrs. J...	female	38	1	0	PC 17599	71.2833	C85
3	1	3	Heikinen, Miss...	female	26	0	0	STON/O2. 3101...	7.925	(NULL)
4	1	1	Futrelle, Mrs. Ja...	female	35	1	0	113803	53.1	C123

SQL Script executing OPENROWSET against <https://calablake141.dfs.core.windows.net/rawdata/titanic.csv> with schema mapping.

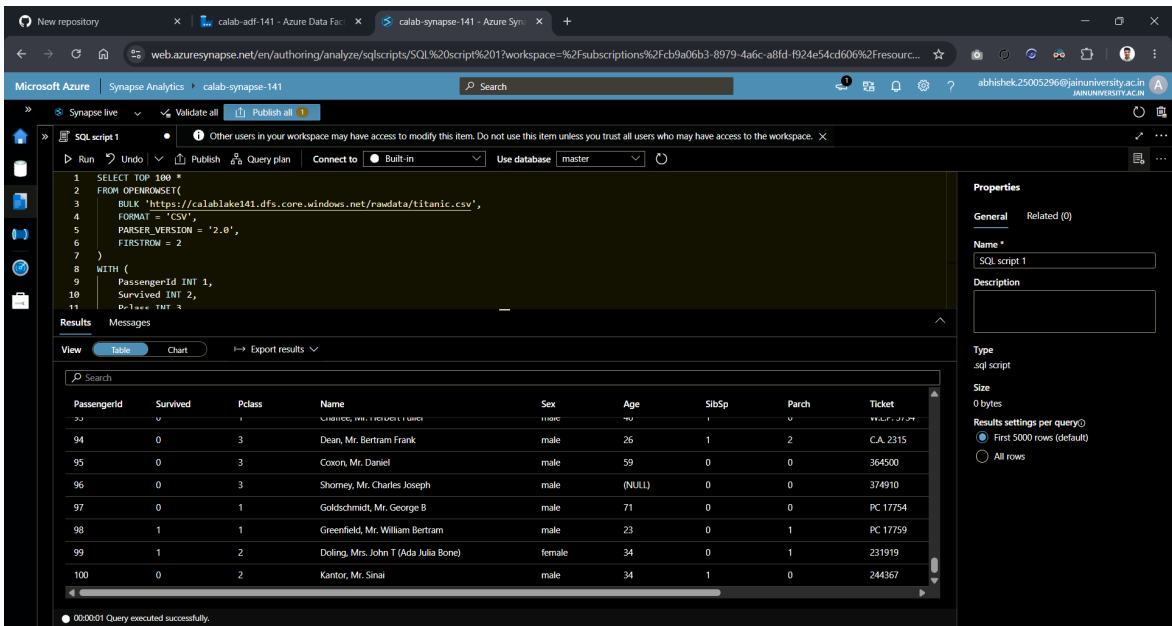
Figure 12: Synapse Query Results & Data Inspection

The screenshot shows the Synapse Query Results & Data Inspection interface. The query is the same as in Figure 11. The results are displayed in a table view with the following columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked. The first four rows of data are shown.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Ow...	male	22	1	0	A/5 21171	7.25	(NULL)	S
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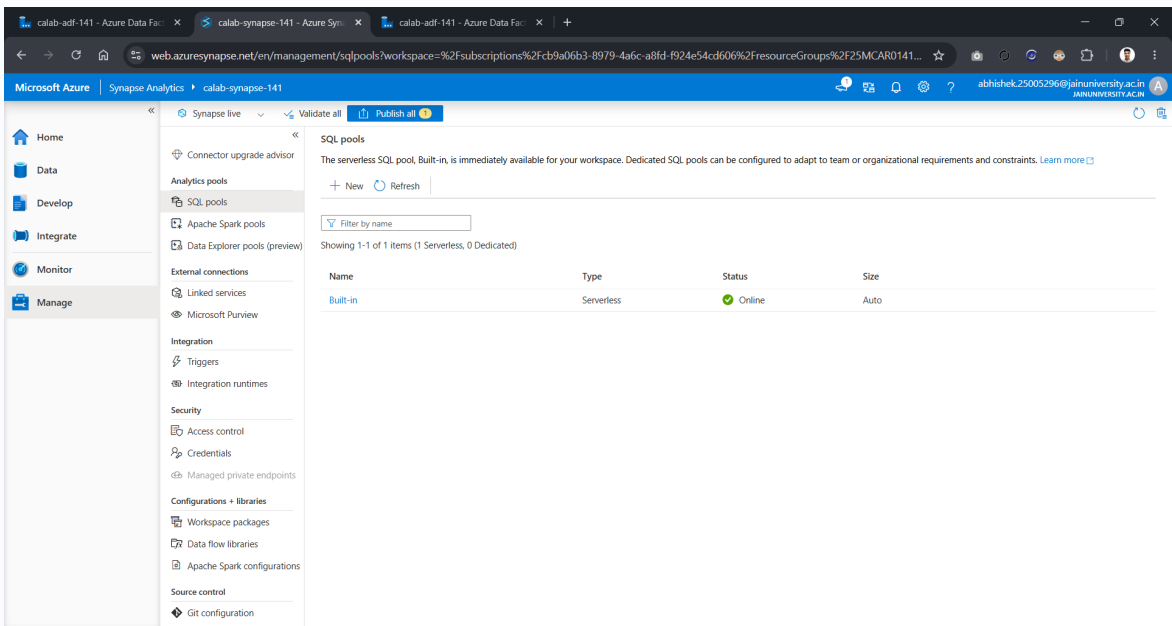
Execution output table displaying queried rows (PassengerId, Survived, Pclass, Name, Sex, Age, Fare, Cabin, Embarked).

Figure 13: Detailed SQL Script Execution & Chart Analysis



Query execution results grid displaying records for Passengers with chart visualization.

Figure 14: Synapse Built-in Serverless SQL Pool Status



Management console showing Built-in Serverless SQL pool running online.

Figure 15: Resource Group Verification

The screenshot shows the Microsoft Azure portal interface. On the left, the 'Resource Manager' sidebar is visible with 'Resource groups' selected. The main content area displays the '25MCAR0141' resource group. The 'Overview' tab is active, showing essential information: Subscription (Azure for Students), Subscription ID (cb9a06b3-8979-4a6c-a8fd-f924e54cd606), and Location (France Central). Below this, a table lists the resources within the group:

Name	Type	Location
calab-adf-141	Data factory (V2)	Central India
calab-synapse-141	Synapse workspace	Central India
calablake141	Storage account	Central India

Final validation of active Azure resources deployed under subscription Azure for Students.

Figure 16: Synapse SQL Request Monitoring Log

The screenshot shows the Microsoft Azure Synapse Analytics 'SQL requests' monitoring dashboard. The dashboard displays a table of SQL requests with the following columns: Request ID, Request content, Submit time, Duration, Data processed, Submitter, and Status. Two requests are listed:

Request ID	Request content	Submit time	Duration	Data processed	Submitter	Status
20670	SELECT TOP 100 * ...	8/10/2026, 1:05:31 PM	0 sec	10.00 MB	abhishek.25005296@jainuniver	Completed
18321	SELECT TOP 10 * F...	8/10/2026, 1:03:46 PM	5 sec	10.00 MB	abhishek.25005296@jainuniver	Completed

Monitoring dashboard showing completed SQL requests, submit times, processing duration, and 10.00 MB data processed.